

POWER MANAGEMENT, CHARGING, AND ENERGY OPTIMIZATION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device incorporates power management architectures configured to support portable operation, safe charging, and efficient energy utilization across diverse deployment environments.

The device may include rechargeable energy storage elements such as lithium-ion batteries, lithium-polymer batteries, solid-state batteries, supercapacitors, or hybrid energy storage systems. Energy capacity may be selected to support single-session operation, extended monitoring, or multi-session usage between charging events.

Charging embodiments may include wired charging interfaces, magnetic connectors, pogo-pin contacts, inductive wireless charging systems, or sealed conductive charging pads configured to maintain device hygiene and fluid resistance.

In certain embodiments, docking stations may provide automatic charging when the device is placed within a storage or calibration dock. Dock-based charging may also enable firmware updates, diagnostic routines, or automated maintenance processes.

Power management circuitry may regulate voltage delivery to sensing components, pumps, processors, communication modules, and safety systems. Dynamic power allocation may prioritize critical sensing operations while reducing consumption of nonessential subsystems.

The computational system may implement sleep modes, low-power standby states, or sensor duty cycling to extend operational lifetime. Sensors may activate only during measurement windows or when motion detection indicates device use.

In some implementations, thermal monitoring systems prevent overheating during charging or operation. Temperature sensors may reduce charging current, disable high-power components, or suspend operation if thermal thresholds are exceeded.

The device may further incorporate battery health monitoring capable of estimating remaining capacity, charge cycles, or long-term degradation. Users may receive notifications recommending replacement or service when battery performance declines.

Energy optimization algorithms may adjust measurement timing, wireless transmission frequency, or inflation control behavior to minimize power consumption without degrading measurement accuracy.

Accordingly, power management and charging embodiments enable reliable portable operation, maintain user safety, and support scalable deployment of the disclosed physiologic measurement system across consumer, research, and clinical-support environments.